

# Molluscan systematics: historical perspectives and the way ahead

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## ABSTRACT

Mollusca, the second-most diverse animal phylum, is estimated to have over 100,000 living species with great genetic and phenotypic diversity, a rich fossil record, and a considerable evolutionary significance. Early work on molluscan systematics was grounded in morphological and anatomical studies. With the transition from oligo gene Sanger sequencing to cutting-edge genomic sequencing technologies, molecular data has been increasingly utilised, providing abundant information for reconstructing the molluscan phylogenetic tree. However, relationships among and within most major lineages of Mollusca have long been contentious, often due to limited genetic markers, insufficient taxon sampling and phylogenetic conflict. Fortunately, remarkable progress in molluscan systematics has been made in recent years, which has shed light on how major molluscan groups have evolved. In this review of molluscan systematics, we first synthesise

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the current understanding of the molluscan Tree of Life at higher taxonomic levels. We then discuss how micromolluscs, which have adult individuals with a body size smaller than 5 mm, offer unique insights into Mollusca's vast diversity and deep phylogeny. Despite recent advancements, our knowledge of molluscan systematics and phylogeny still needs refinement. Further advancements in molluscan systematics will arise from integrating comprehensive data sets, including genome-scale data, exceptional fossils, and digital morphological data (including internal structures). Enhanced access to these data sets, combined with increased collaboration among morphologists, palaeontologists, evolutionary developmental biologists, and molecular phylogeneticists, will significantly advance this field.

**Key words:** Mollusca, micromolluscs, taxonomy, phylogeny, systematics.

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## I. INTRODUCTION

Molluscs, recognised for their economic importance as sources of food and jewellery and as model organisms in neurobiology (Katz, 2009; Kocot *et al.*, 2011; Ponder, Lindberg & Ponder, 2019), are renowned for their exceptional diversity and rich fossil record. They constitute the second-largest animal phylum, with an estimated 100,000 living species (Parkhaev, 2008, 2017; Ponder & Lindberg, 2008; Kocot, 2013). Molluscs inhabit a wide range of environments, ranging from aquatic to terrestrial habitats, and extend to extreme locales such as deep-sea hydrothermal vents and cold seeps (Lemer, Bieler & Giribet, 2019; Sun *et al.*, 2020). Molluscs have evolved physiological and morphological adaptations and developed diverse lifestyles, including crawling, free-swimming, and engaging in symbiotic relationships (Distel *et al.*, 2017; Peijnenburg *et al.*, 2020; Guo *et al.*, 2023). Although some molluscs are categorised as biofoulers and invasive species (Frey *et al.*, 2014; Yang *et al.*, 2018), others are threatened or endangered (Ferreira-Rodriguez *et al.*, 2019; Barord *et al.*, 2023), and many serve as valuable indicators of climate change, environmental shifts, and marine biodiversity (Middelfart, Kirkendale & Wilson, 2016; Peijnenburg *et al.*, 2020). The fossil record suggests that Mollusca emerged around 542–530 million years ago (Parkhaev & Demidenko, 2010; Ponder *et al.*, 2019), positioning them at the forefront of the Cambrian Explosion and making them a pivotal group for studying early Bilaterian evolution, due to their high preservation potential (Wang *et al.*, 2017; Huan *et al.*, 2020; Wang *et al.*, 2020; Han *et al.*, 2022).

The existing eight classes of molluscs encompass a vast array of life forms: the diverse Gastropoda, including snails, abalones, slugs, and limpets; the tubular Scaphopoda, known as tusk shells; the filter-feeding Bivalvia, such as clams, scallops, oysters, and mussels; the intelligent Cephalopoda, comprising octopuses, squids, cuttlefishes, and nautilus; the limpet-like Monoplacophora; the worm-like Solenogastres, or Neomeniomorpha; the burrowing Caudofoveata, also known as Chaetodermomorpha; and the armour-plated Polyplacophora, or chitons (Fig. 1). There are several alternative hypotheses regarding the relationships among the molluscan classes (Fig. 2).

In the last two decades, the field of molluscan molecular systematics has made significant progress, transitioning from the analysis of a limited number of gene markers (Winnepeninckx, Backeljau & DeWachter, 1996; Passamanek, Schander & Halanych, 2004; Giribet *et al.*, 2006; Meyer *et al.*, 2010; Wilson, Rouse & Giribet, 2010) to employing comprehensive data sets encompassing hundreds of nuclear genes (Kocot *et al.*, 2011, 2019, 2020; Smith *et al.*, 2011; Tanner *et al.*, 2017; Lemer *et al.*, 2019; Cunha, Reimer & Giribet, 2022; Song *et al.*, 2023). This advancement has significantly enhanced our understanding of the relationships within and among major lineages. However, numerous uncertainties still persist within certain clades. Although phylogenetic relationships of the eight mollusc classes have recently

become clearer, additional effort on ordinal-level relationships would help improve understanding of molluscan evolutionary history (Fig. 3).

The taxon coverage of genomic data in molluscs is uneven, with a priority given to taxa of greater relevance to human interests and economic significance (Sigwart *et al.*, 2021). For example, *Magallana gigas* (the Pacific oyster) has undergone genome-scale sequencing (Zhang *et al.*, 2012). As of November 10, 2023, data from the National Center for Biotechnology Information (NCBI, <https://www.ncbi.nlm.nih.gov/>) indicate that gastropods and bivalves constitute 42.23% and 41.55% of sequenced molluscan genomes, respectively. By contrast, genomic data for other molluscan classes comprise less than 10%, with Monoplacophora still lacking genome data (Fig. 4). Additional sequencing data remains essential for some less-common groups (Song *et al.*, 2023; Wang *et al.*, 2024b; Fig. 4). Notably, micromolluscs, typically with adult sizes of 5 mm or less, have been overlooked despite mostly being gastropods and bivalves (Geiger *et al.*, 2007; Sasaki, 2008; Albano *et al.*, 2021). Their rich species diversity has contributed to elucidating deeper molluscan relationships (Varney *et al.*, 2021). Our understanding of molluscan nuclear genomes continues to grow. Molluscan genomes have provided essential resources, as transcriptomes and genomes can provide a wealth of (predicted) gene identities for phylogenomic analyses (Smith *et al.*, 2011; Zapata *et al.*, 2014). Additionally, fossils and shells serve as valuable sources of molecular data, applicable in stratigraphic and molecular dating techniques (Geist, Wunderlich & Kuehn, 2008; Andree & López, 2013; Hawk & Geller, 2019; Ponder *et al.*, 2019; Li *et al.*, 2023).

To promote an improved understanding of their evolution, we provide a historical perspective of molluscan systematics and a comprehensive overview of the latest developments in this field. We examine the current knowledge of molluscan phylogeny, focusing on relationships at both classes and other higher levels (e.g. orders). We then explore the significance of micromolluscs in elucidating the remarkable diversity and deep relationships within Mollusca, thus bridging gaps between well-studied groups and taxa that remain yet to be discovered. In addition to high-quality genomes and expanded sampling of taxa, the development of molluscan systematics will be further propelled by more comprehensive data, including fossil records and digital morphological data. Additionally, collaboration between molecular systematists and morphologists will play a crucial role in advancing the field.

## II. THE DEEP PHYLOGENY OF MOLLUSCS

Phylogenetic hypotheses for the major molluscan classes were initially reconstructed on the basis of morphological characters such as shell structure, and morphology of the mantle, gut, and nervous system (Runnegar & Pojeta, 1974; Salvini-Plawen, 1985; Scheltema, 1993, 1996; Ivanov, 1996; Waller, 1998; Haszprunar, 2000; Fig. 2). However,



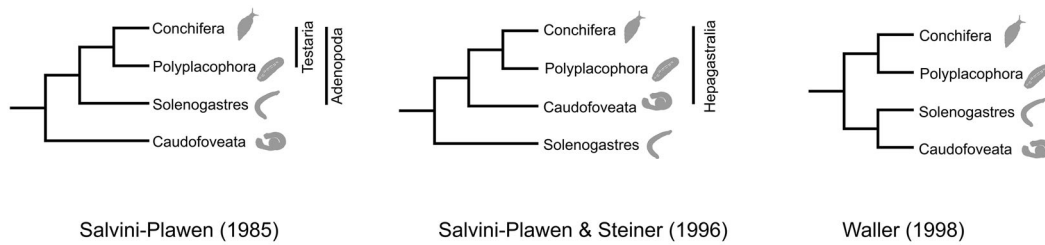
**Fig. 1.** Diversity of extant molluscs. (A) *Epimenia* sp. (Solenogastres); (B) *Chaetoderma* sp. (Caudofoveata); (C) *Callochiton puniceus* (Polyplacophora); (D) *Laevipilina* sp. (Monoplacophora); (E) *Nautilus macromphalus* (Cephalopoda); (F) *Callistoctopus* sp. (Cephalopoda); (G) *Xipholeptos notoides* (Cephalopoda); (H) *Todarodes pacificus* (Cephalopoda); (I) *Siphonodentalium dalli* (Scaphopoda); (J) *Amphidromus atricallosus* (Gastropoda); (K) *Goniobranchus sinensis* (Gastropoda); (L) *Cabestana spengleri* (Gastropoda); (M) *Potamilus fragilis* (Bivalvia); (N) *Argopecten irradians* (Bivalvia); (O) *Tridacna crocea* (Bivalvia); (P) *Carinaria cristata* (Gastropoda). Images A, B, and D are courtesy of Kevin M. Kocot. Image I is courtesy of Nickellaus Roberts. Image K is credited to Josy Lai. Images H and O are credited to Albert Kang. Images C, E–G, J, L, M, and P are credited to Yifan Zou, Shankar Meyer, itai Grisaru, Ron Greer, klearad, Felix Harper, pocahontas7780, Neil McDaniel, respectively. Images C, E–H, J–M, O, and P are used under CC BY-NC 4.0 (<https://creativecommons.org/licenses/by-nc/4.0/>). Image N is courtesy of Biyang Xu. All images were slightly cropped.

relying solely on morphological characters is less successful when investigating the deeper relationships within Mollusca, especially for discerning derived *versus* ancestral (plesiomorphic) traits, identifying homologous structures, and considering the effects of heterochrony (Kocot *et al.*, 2011; Kocot, Halanych & Krug, 2013; Stöger *et al.*, 2013). DNA-based phylogenetics has substantially improved our understanding of the molluscan Tree of Life (ToL) (Kapli, Yang & Telford, 2020). In the last two decades, such methods have been used to reconstruct the relationships of major molluscan lineages, evolving from oligo genes to genome-wide data, and integrating this with morphological and fossil

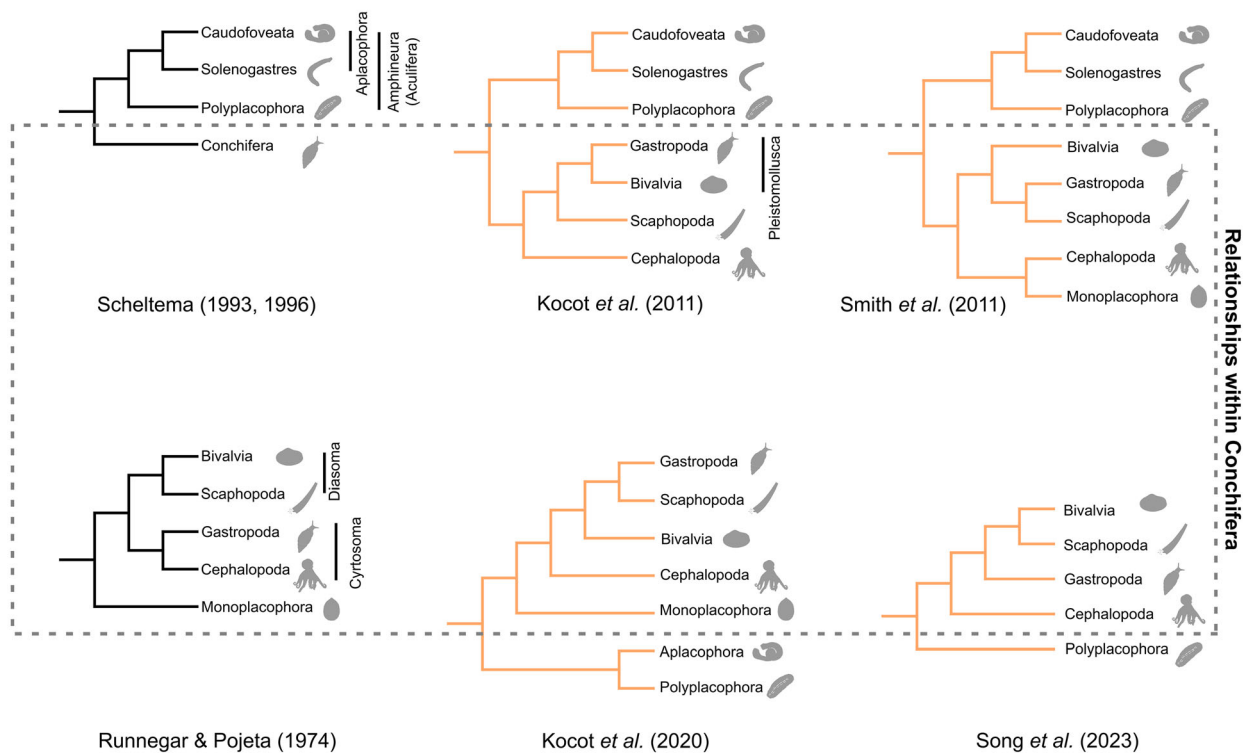
information (Passamanek *et al.*, 2004; Giribet *et al.*, 2006; Meyer *et al.*, 2010; Wilson *et al.*, 2010; Kocot *et al.*, 2011, 2019, 2020; Smith *et al.*, 2011; Kano *et al.*, 2012; Stöger *et al.*, 2013; Song *et al.*, 2023; Fig. 2).

In the early 2000s, reconstructing the molluscan ToL using molecular data was in its infancy. Due to the technology available at the time, deep molluscan phylogenetics was inferred mainly based on limited gene markers [small subunit (SSU) and/or large subunit (LSU) ribosomal RNA (rRNA)] (Steiner & Dreyer, 2003; Passamanek *et al.*, 2004; Giribet *et al.*, 2006; Wilson *et al.*, 2010). Boore, Medina & Rosenberg (2004) and Dreyer & Steiner (2004) attempted to reconstruct

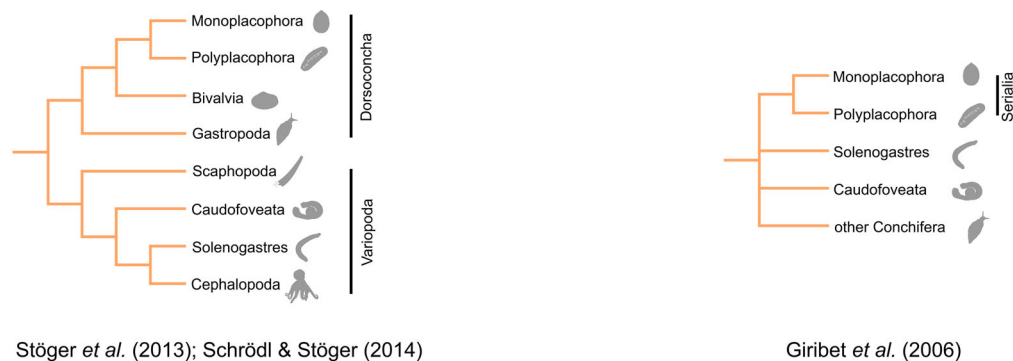
### A Scenarios for the Testaria concept



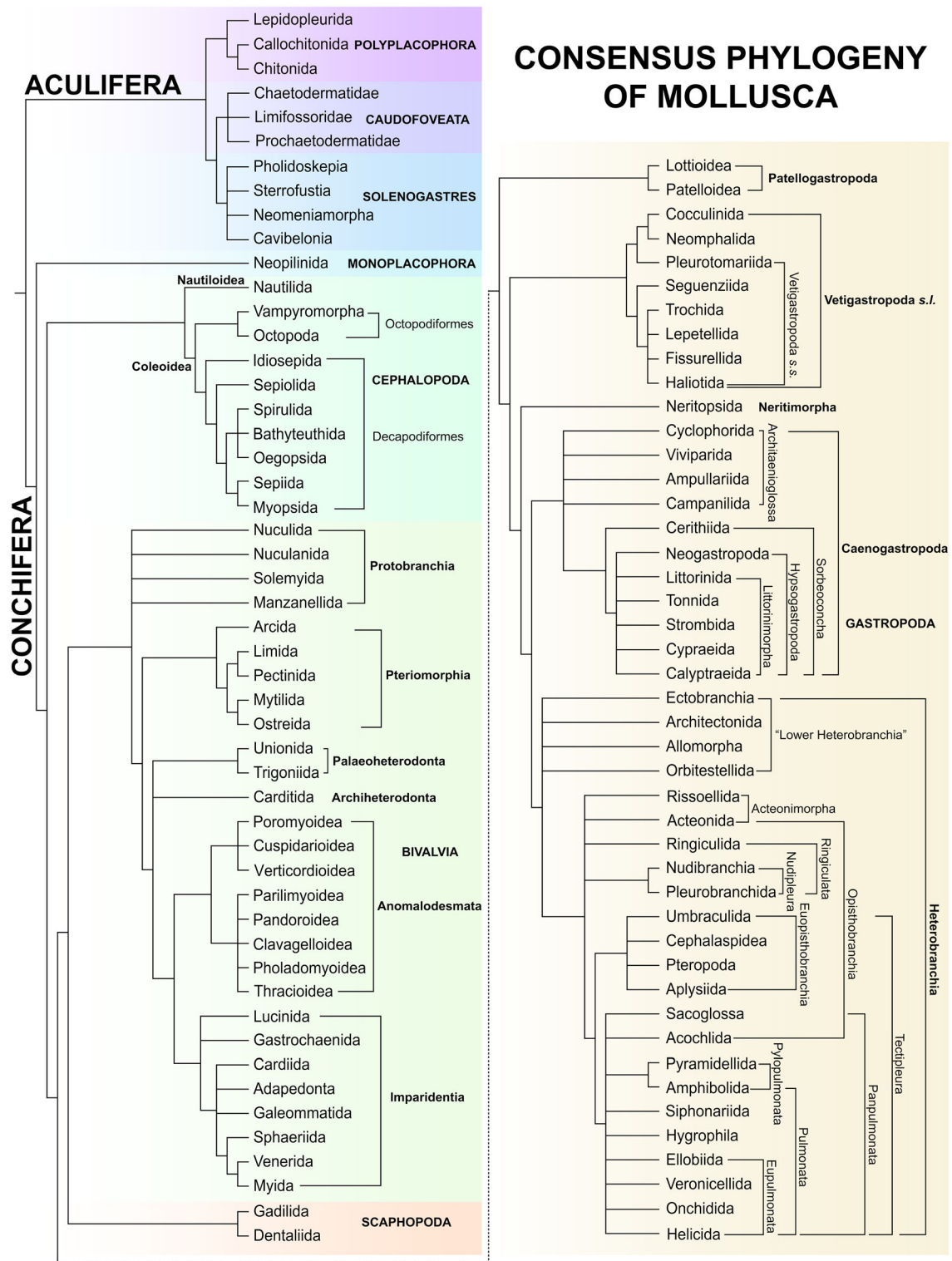
### B Monophyletic Aculifera as sister to Conchifera



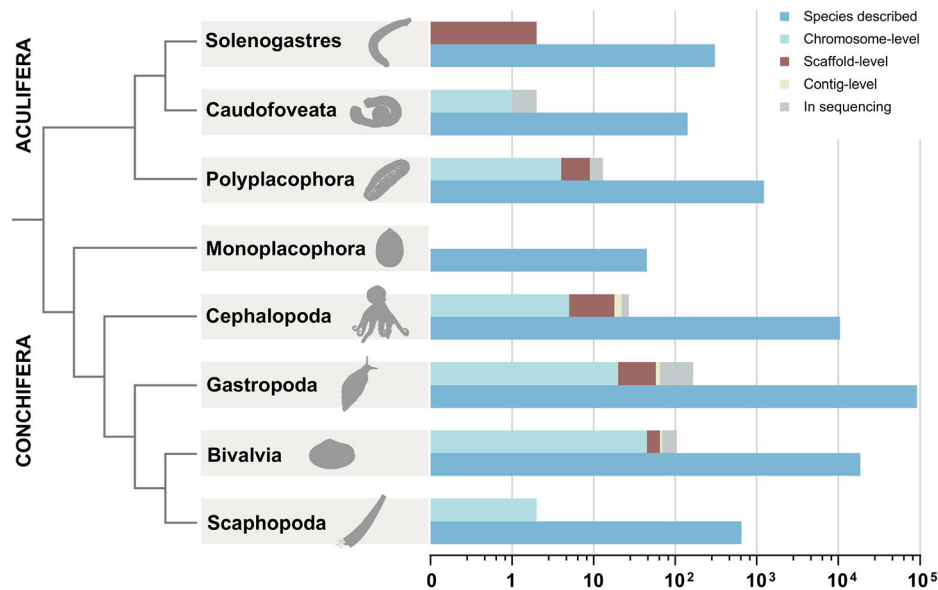
### C Variopoda, Dorsoconcha, and Serialia



**Fig. 2.** Alternative hypotheses of relationships among major living molluscan lineages, with molecular analyses highlighted in orange. The relationships within Conchifera are displayed inside the dashed box. The Serialia hypothesis is not widely accepted, likely due to the chimeric 28S rRNA sequence and the largely unresolved remaining nodes within the Mollusca (Wanninger & Wollesen, 2019).



**Fig. 3.** Consensus phylogeny among major lineages of living molluscs. Relationships that cannot be explained by morphological characters, that show conflicts among studies, or that have minimal support are depicted as polytomies. The selection criteria for the studies are as follows: (i) classic morphological studies; (ii) molecular studies with the broadest taxon sampling; and (iii) phylogenomic studies with the most extensive taxon sampling possible. References used to assemble the phylogeny are marked with an asterisk (\*) in the reference list.



**Fig. 4.** Cladogram for lineages of Mollusca with the number of described species available and genome information. The blue bar indicates the estimated numbers of described living species obtained from the Catalog of Life. Stacked bars to the right indicate the numbers of complete genomes at different assembly levels and number of species prepared for genome sequencing as of August 15, 2024. Contig-level = the fundamental units of genome assembly, created through the overlap of sequencing reads. Scaffold-level = longer structures assembled by grouping overlapped contigs, approximating the authentic structure of chromosomes. Chromosome-level = scaffolds further assembled to reconstruct the complete chromosome structure, filling in unknown regions between them. Typically, each chromosome comprises one or more scaffolds. The original organism silhouettes for Polyplacophora and Solenogastres are credited to Noah Schlottman and are used under CC BY-SA 3.0 (<https://creativecommons.org/licenses/by-sa/3.0/>) with colour changed. The silhouette of Caudovoate is built upon the image credited to Elena Gerasimova under CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0/>) with colour changed.

molluscan phylogenetics based on mitochondrial protein-coding genes, but with relatively limited taxon sampling. Consequently, these early studies, primarily utilising ribosomal and mitochondrial protein genes, yielded low-resolution insights into the divergence of the earliest molluscan branches, with two significant shortcomings: a lack of robust support for deep-branching nodes and persistent issues with the non-monophyly of major taxa including Bivalvia, Gastropoda, and the phylum Mollusca itself. High-throughput sequencing technology has revolutionised molecular systematics (Metzker, 2010), profoundly changing our understanding of the evolutionary history of Mollusca. The extant molluscan classes have been investigated in detail over the past 15 years (Haszprunar, Schander & Halanych, 2008), yet reconstruction of the molluscan ToL remains challenging (Stöger *et al.*, 2013).

The placement of Solenogastre, Caudovoate, and Polyplacophora remains a topic of ongoing discussion and has given rise to two contradicting hypotheses (Salvini-Plawen & Steiner, 2014). Of these, Testaria, a monophyletic clade ally-ing living Polyplacophora with Conchifera, is supported by distinct morphological synapomorphies like the radula complex, the division of the midgut, dorsoventral musculature, and excretory organs (Salvini-Plawen & Steiner, 2014). However, the absence of a true periostracum and a differentiated shell gland in chitons is at odds with their proposed

close relationship with conchiferans (Kocot, 2013). Aculifera, a clade that unites Aplacophora and Polyplacophora and is then sister to Conchifera (all other classes), is supported not only by characteristics such as sclerites and epidermal papillae (Scheltema, 1993, 1996), but also by most phylogenomic studies (Kocot *et al.*, 2011, 2017, 2020; Smith *et al.*, 2011; Vinther *et al.*, 2012; Fig. 2). Considering the structural similarities of aplacophoran and chiton sclerites, which are distinct from those of conchiferans, along with the role of the basal cell in the secretion process (Kocot, 2013), an aculiferan–conchiferan dichotomy provides a valuable phylogenetic framework for further exploration of the homology of aculiferan sclerites. Alternatively, the hypothesised Serialia (Monoplacophora + Polyplacophora; Giribet *et al.*, 2006) has been refuted by phylogenetic statistical testing (Smith *et al.*, 2011; Kocot *et al.*, 2020; Fig. 2), even though individuals within this grouping all possess serially repeated ctenidia and eight sets of shell muscles. While the hypotheses raised (for Testaria, Aculifera, and Serialia) have some support from morphology, there remain other traits that conflict within specific hypotheses, demonstrating the limitations of phylogenetic construction based on morphology alone.

Additionally, discrepancies persist within Conchifera (Fig. 2). Monoplacophora, which live exclusively in deep waters and have a rich fossil record, have long been considered a separate class with Early Cambrian limpet-like affinities

(Knight *et al.*, 1960). Phylogenetic placement of Monoplacophora is variable among molecular studies (Smith *et al.*, 2011; Kocot *et al.*, 2020). Smith *et al.* (2011) conducted the first molluscan phylogenomic study that included the monoplacophoran *Laevipilina hyaline*. This study recovered a clade uniting Monoplacophora and Cephalopoda with strong support. This relationship has also been supported by the presence of multi-septate shells in both fossil monoplacophorans and cephalopods (Yochelson, Flower & Webers, 1973). In addition, with genomic data from the monoplacophoran *Laevipilina antarctica*, a sister-group relationship between Monoplacophora and Cephalopoda was recovered in RAxML (randomised accelerated maximum likelihood)-based maximum likelihood analysis but only with moderate support (Kocot *et al.*, 2020). Despite the fossil record and morphological features suggesting a close evolutionary relationship between these two taxa, which differ dramatically in appearance (Yochelson *et al.*, 1973; Holland, 1987; Kröeger, Vinther & Fuchs, 2011), placement of Monoplacophora as sister to all other conchiferans was recovered in most analyses conducted by Kocot *et al.* (2020). Therefore, the early branching of Monoplacophora within Conchifera indicates that the reduction of dorsoventral muscle pairs is an apomorphy of a clade of true-shell molluscs (Kocot *et al.*, 2020).

The relationships among Scaphopoda, Gastropoda, and Bivalvia (i.e. Ganglionata, reviewed by Schrödl & Stöger, 2014) have long been unstable across studies and remain contentious (Kocot *et al.*, 2011, 2020; Smith *et al.*, 2011; Vinther *et al.*, 2012; Cunha & Giribet, 2019; Song *et al.*, 2023; Sigwart *et al.*, 2024; Fig. 2). Two lobes in the ontogeny of scaphopod larval shell formation suggest a close relationship between Scaphopoda and Bivalvia (i.e. Diasoma; Runnegar & Pojeta, 1974; Fig. 2). This clade was not recovered by most phylogenomic studies (Kocot *et al.*, 2011, 2020; Smith *et al.*, 2011), but was recovered in coalescent-based trees (Cunha & Giribet, 2019; Sigwart *et al.*, 2024) and in whole-genome-based phylogenomics (Song *et al.*, 2023; Sigwart *et al.*, 2024). Kocot *et al.* (2011) proposed Pleistomollusca for uniting Gastropoda + Bivalvia, given strong support for this clade and features in common, such as the shared veliger larval stage and loss of the anterior ciliary rootlet in locomotory cilia (Lundin, Schander & Todt, 2009). Alternatively, a sister relationship between Scaphopoda and Gastropoda was also recovered (Smith *et al.*, 2011; Cunha & Giribet, 2019; Kocot *et al.*, 2020; Sigwart *et al.*, 2024). However, no clear morphological synapomorphies were detected for this clade (Kocot *et al.*, 2020). This highlights the internal inconsistencies within Conchifera, which make it difficult to assess the character states of the last common conchiferan ancestor (Wanninger & Wollesen, 2019). A comprehensive class-level phylogenomic reconstruction in molluscs is critical for accurately interpreting shared biological traits, and determining whether they are ancestral (symplesiomorphies), derived (synapomorphies), or the result of convergent evolution.

While the shell characters of the last common ancestor of extant molluscs are still controversial, a combination of early fossils and the phylogenetic framework of living molluscs

could allow us to deduce the morphology of their ancestors (Kocot *et al.*, 2020; Zhang *et al.*, 2024). Studying molluscan fossil taxa can reveal clues to the timing and features of ancient molluscs, but the classification of some fossils is still disputed. For example, the Ediacaran fossil *Kimberella* has been described variously as a bilaterian (Erwin & Davidson, 2002) or more like a mollusc (Fedonkin & Waggoner, 1997), and even showing affiliation with a Cambrian halkieriid [a presumed stem aculiferan (Seilacher, 1999; Vinther, 2009)]. Other fossils, such as the Middle Cambrian *Odontogriphus* and *Wiwaxia*, which possess mouthparts with an affinity to the molluscan radula, might provide insights into the original form of the molluscan radula (Smith, 2012). Given that *Kimberella*, from the Ediacaran, is widely accepted as the oldest bilaterian fossil, pre-dating most other putative mollusc fossils from the Cambrian (including *Wiwaxia*, *Halkieria*, and hyoliths) (Smith, 2014; Li, Skovsted & Topper, 2022), the accurate classification of *Kimberella* will not only be crucial for identification of the earliest molluscs, but also for understanding the Cambrian explosion.

### III. RELATIONSHIPS WITHIN MAJOR LINEAGES

#### (1) Relationships within Bivalvia

The interrelationships of the modern bivalve higher groups have been a subject of heated debate in the past decade. Various methods have been used to inform these discussions, including morphology (e.g. shell hinge, muscle scars, and structures of the gills and stomach), Sanger-based sequencing [e.g. Bieler *et al.*, 2014 (with morphological data); Combosch *et al.*, 2017], phylotranscriptomics (González *et al.*, 2015; Lemer *et al.*, 2019), and analyses of ultraconserved elements (UCEs) (González-Delgado, Rodríguez-Flores & Giribet, 2024). Recently, a refined classification for living bivalves has been adopted, including six major higher taxa: Protobranchia, Pteriomorphia, Archiheterodonta, Palaeoheterodonta, Anomalodesmata, and Imparidentia (Bieler *et al.*, 2014) (Fig. 3).

##### (a) Protobranchia

Protobranchia is a gill-based concept, with gills used primarily for respiration, and is commonly recognised as the most ancient divergent subclass of Bivalvia. Within Protobranchia, three orders were traditionally recognised – Nuculida, Nuculanida, and Solemyida – based on ciliary sorting areas and typhlosole (Villarroel & Stuardo, 1998; Sharma *et al.*, 2013). Ponder *et al.* (2019) reassigned the superfamily Manzanelloidea from Solemyida to a new order, Manzanellida. Although Manzanellida has not been adopted yet in the current ranked bivalve classification (MolluscaBase, 2024), the homogeneous structure of inconspicuous granular crystals and meaningful bootstrap value do support this change (Sato *et al.*, 2020).

Relationships among these orders vary across studies. Protobranch non-monophyly was reported from studies using Sanger-based sequencing (e.g. Bieler *et al.*, 2014; Combosch *et al.*, 2017) and analyses based on UCEs (González-Delgado *et al.*, 2024). However, both several-gene analyses (Sharma *et al.*, 2012; Bieler *et al.*, 2014; Sato *et al.*, 2020) and phylogenomic approaches have recovered monophyly of Protobranchia (Kocot *et al.*, 2011; Smith *et al.*, 2011; González *et al.*, 2015; Lemer *et al.*, 2019). Thus, monophyly of Protobranchia remains unstable (Fig. 3), with incongruences hard to arbitrate (Lemer *et al.*, 2019). Increasing the sampling of these ancient deep-water bivalves could help resolve these relationships (Lemer *et al.*, 2019; González-Delgado *et al.*, 2024).

Although shell microstructures are useful to classify living protobranchs (Zardus, 2002; Bieler *et al.*, 2014; Sato *et al.*, 2020) and some subtle shell microstructural differences have potential systematic value for identifying nukulids (Sato *et al.*, 2020), significant genetic divergence within the morphological species of Protobranchia (Neulinger *et al.*, 2006; Zardus *et al.*, 2006) implies that many cryptic protobranchs await sampling, description, and phylogenetic classification.

#### (b) *Autobranchia*

All other bivalves are grouped into Autobranchia, which is defined by having enlarged gills used for both filter-feeding and respiration, distinct from the protobranch type. Autobranchia is composed of Pteriomorphia and Heteroconchia [which contains Palaeoheterodonta + Heterodonta (Archiheterodonta + Euheterodonta (Anomalodesmata + Imparidentia))] (Giribet & Wheeler, 2002; Bieler *et al.*, 2014). The monophyly of Autobranchia is mostly stable based on anatomical features (e.g. Giribet & Wheeler, 2002) and molecular studies (e.g. Bieler *et al.*, 2014; González *et al.*, 2015; Lemer *et al.*, 2019), but within this clade, competing ideas remain, involving: (i) the relationships within Heteroconchia; and (ii) the internal relationships within Pteriomorphia and Euheterodonta.

Traditionally, Palaeoheterodonta was placed as the earliest branching heteroconchian lineage (Waller, 1998; Giribet & Wheeler, 2002) or recovered as a non-monophyletic group (Purchon, 1987; Salvini-Plawen & Steiner, 1996). The traditional hypothesis separating Palaeoheterodonta from Heterodonta and the internal resolution of the main Heteroconchian lineages has received strong support in recent phylogenomic studies (González *et al.*, 2015; Lemer *et al.*, 2019). However, some molecular analyses have recovered Archiheterodonta or a clade uniting Archiheterodonta and Palaeoheterodonta as the earliest-branching lineage (Wilson *et al.*, 2010; Sharma *et al.*, 2012; Bieler *et al.*, 2014; González-Delgado *et al.*, 2024). These incongruences make the base of Heteroconchia unstable.

#### (c) *Pteriomorphia*

Monophyly of Pteriomorphia has been corroborated by Sanger-based sequencing (e.g. Steiner & Hammer, 2000;

Giribet & Wheeler, 2002) and phylogenomics (González *et al.*, 2015; Lemer *et al.*, 2016, 2019). Five extant orders are now recognised: Arcida, Limida, Mytilida, Ostreida and Pectinida (Bieler *et al.*, 2014; Lemer *et al.*, 2016). However, the internal relationships within this early Ordovician group have long been uncertain (Waller, 1998; Steiner & Hammer, 2000; Giribet & Wheeler, 2002; Bieler *et al.*, 2014).

The placement of Mytilida and Arcida is a central topic of debate. Elucidating their relative positions will contribute to understanding the distribution and evolution of calcite and nacre within pteriomorphians and their relatives (e.g. Lemer *et al.*, 2016). For many years, both Mytilida and Arcida were considered candidates for the earliest offshoots of Pteriomorphia (e.g. Carter, 1990; Waller, 1998; Lemer *et al.*, 2016). Notably, recent phylogenomics analyses consistently favour Arcida as the earliest branch of pteriomorphs (Lemer *et al.*, 2016, 2019). Although another study based on UCEs found Arcida and Mytilida to be sister groups (González-Delgado *et al.*, 2024), placement of Mytilida as the sister group to Ostreida is currently considered the best-supported hypothesis (Lemer *et al.*, 2016). The early branching of Arcida implies that the outer calcific shell layer found in other pteriomorphians excluding Arcida (Esteban-Delgado *et al.*, 2008; Bieler *et al.*, 2014), which likely is an adaptation to resist shell dissolution (Taylor & Reid, 1990; Harper, 2000), was an important innovation during their adaptive radiation (Lemer *et al.*, 2016).

#### (d) *Anomalodesmata*

A sister relationship between Anomalodesmata and Imparidentia has been well supported (Bieler *et al.*, 2014; Combosch *et al.*, 2017; Lemer *et al.*, 2019; González-Delgado *et al.*, 2024). Within the Anomalodesmata, previous hypothesis suggested a deep subdivision of three main clades: ‘Septibranchia’ (comprising Verticordiidae, Euciroidae, Poromyidae, and Cuspidariidae), ‘lyonsiids’ (Lyonsiidae, Clavagellidae, Pandoridae, and Laternulidae), and ‘thraciids’ (Thraciidae, Cleidothaeridae, Myochamidae, and Periplomatidae). The latter two clades were largely congruent across Sanger-based sequencing studies (Harper, Dreyer & Steiner, 2006; Bieler *et al.*, 2014; Combosch *et al.*, 2017). Of these, placement of septibranchs was unstable across molecular analyses. It was recovered as the sister group to the other anomalodesmatans, albeit with a lack of node resolution (Bieler *et al.*, 2014). A later phylogenomics study (Lemer *et al.*, 2019), largely recovered this group as sister to the ‘thraciids’. However, monophyly of the ‘Septibranchia’ has been questioned (Harper *et al.*, 2006; Combosch *et al.*, 2017).

Currently, a deep dichotomy is represented by two distinct clades, a non-carnivorous lineage and a carnivorous lineage. This division has been well supported by an extensive morphological phylogenetic analysis, which noted features such as the presence of a plicate ctenidium in the former and an enhanced digestive system in the latter (Machado & Passos, 2022). It was also corroborated by a phylogenomic study with the largest sampling to date (González-Delgado *et al.*, 2024). However,

divisions for families or superfamilies (e.g. Pandoroidea) within the non-carnivorous lineage are not congruent between molecular-based (Williams *et al.*, 2017; González-Delgado *et al.*, 2024) and morphological studies (Machado & Passos, 2022). Consequently, relationships among anomalodesmatan superfamilies are in flux (Fig. 3).

An insufficient number of loci and sparse sampling represent obstacles to resolving the internal relationships within Anomalodesmata in phylogenetic studies (Machado & Passos, 2022; González-Delgado *et al.*, 2024). Adding more deep-sea families of Anomalodesmata would be a logical next step (González-Delgado *et al.*, 2024). Additionally, the classification of Anomalodesmata itself needs to be reassessed (Ponder *et al.*, 2019).

#### (e) *Imparidentia*

Imparidentia underwent rapid radiation and this clade is diverse, spanning marine, brackish, and freshwater taxa. It was first named by Bieler *et al.* (2014). Clarifying the relationships within Imparidentia has prove challenging both with Sanger-based markers and morphological methods (e.g. Giribet & Wheeler, 2002; Bieler *et al.*, 2014). Phylogenomics provides the potential for elucidating the relationships within Imparidentia; however, there are long-debated issues regarding: (i) the deepest branching pattern of Lucinida and Gastrochaenidae; (ii) the position of Adapedonta, Galeommatida, and Cardiida; and (iii) improving the resolution within Neoheterodonte [a subclade of Imparidentia comprising Sphaeriidae, Myida, and Venerida], particularly the problematic position of Chamidae within Venerida.

Based on molecular evidence, Lucinidae and Thyasiridae are the first offshoots of Imparidentia (Lemer *et al.*, 2019; González-Delgado *et al.*, 2024). This challenges the traditional view placing Thyasiridae as a member of Lucinida, based on morphological similarities including the general shell shape, an extensible foot, and thickened ctenidia with a bacteriocyte zone (Southward, 1986), and thus renders Lucinida paraphyletic. Whether Lucinida, Thyasiridae, and Gastrochaenidae form a grade remains elusive. Additionally, the next-branching group of Adapedonta, Galeommatida and Cardiida was unstable before a well-resolved Neoheterodonte was identified (González *et al.*, 2015; Lemer *et al.*, 2019; González-Delgado *et al.*, 2024). Within Neoheterodonte, despite a sister relationship between Myida and Venerida being compatible with both phylogenomics (Lemer *et al.*, 2019; González-Delgado *et al.*, 2024) and mitogenome-based approaches (Wang *et al.*, 2023), relationships within Venerida remain uncertain, especially the position of the highly autapomorphic Chamidae, even with increased sampling of Chamoidea (Lemer *et al.*, 2019; Wang *et al.*, 2023). UCE probes have been highly informative at the species level for imparidentian bivalves (González-Delgado *et al.*, 2024), although unfortunately, no representatives of Chamidae were explored.

Although phylogenomic-based methods still need to improve in terms of taxonomic sampling, they have provided

new insights regarding the classification of Imparidentia: (i) re-evaluation of Lucinida; (ii) elevation of the three superfamilies, Gastrochaenoidea, Galeommatoidea, and Sphaerioidea to the ordinal level; (iii) the validity of some superfamilies, such as Arcticoidea and Glossoidea; and (iv) the recognition of Mactroidea, Ungulinoidea, and Chamoidea as members of Venerida (Lemer *et al.*, 2019).

## (2) Relationships within gastropods

### (a) *Backbone for Gastropoda*

Gastropoda is a class of molluscs that is not only morphologically highly diverse, but is also the second most diverse among metazoan classes (Ponder *et al.*, 2019; Uribe *et al.*, 2022). Generally, gastropods can be identified by unique morphological features, including a single shell, an operculum, and the occurrence of torsion during larval development (Zapata *et al.*, 2014). The living gastropods include the Patellogastropoda, Vetigastropoda *s.l.*, Neritimorpha, Caenogastropoda, and Heterobranchia (Uribe *et al.*, 2022) (Fig. 3). Notably, challenges remain in reconstructing a robust phylogeny of gastropods, thus placing it at the centre of debates in molluscan systematics (Zapata *et al.*, 2014; Uribe *et al.*, 2022).

Previous hypotheses on the evolution of gastropods were constructed using characteristics from morphology, anatomy, and development (Ponder & Lindberg, 1997; Sasaki, 1998). These hypotheses agree with a dichotomy between Eogastropoda (Patellogastropoda) and Orthogastropoda (which includes the rest of the gastropods that possess features such as a flexoglossate radula and a single left hypobranchial gland). With the advance of sequencing technologies and better sampling across lineages, the understanding of evolutionary relationships among the main lineages and their circumscription improved. A sister relationship between Apogastropoda (Caenogastropoda + Heterobranchia) and Neritimorpha was congruent across morphological, mitogenomic and nuclear phylogenomic approaches (Ponder & Lindberg, 1997; Kocot *et al.*, 2011; Simone, 2011; Zapata *et al.*, 2014; Cunha & Giribet, 2019; Uribe *et al.*, 2019, 2022).

However, incongruences exist among phylogenomics studies regarding the rooting of the gastropod tree (Zapata *et al.*, 2014; Cunha & Giribet, 2019; Chen & Schrödl, 2022; Uribe *et al.*, 2022). Patellogastropoda was grouped into a clade with Vetigastropoda *s.s.* [Psilogastropoda (Zapata *et al.*, 2014; Cunha & Giribet, 2019)], and then proposed as a sister group to Angiogastropoda [Neritimorpha (Caenogastropoda + Heterobranchia)]. This proposal received strong support, and shared features respectively include the presence of unprotected gametes and an enclosed embryo (see Section III.2.d) (Cunha & Giribet, 2019). In another scenario, Patellogastropoda was recovered as a sister group to Vetigastropoda + Neomphalina when the number of single-copy orthologs was increased (Chen & Schrödl, 2022). Consequently,

these three studies rejected the Orthogastropoda hypothesis. By contrast, when two deep-sea clades (Neomphalida and Cocculinida) were incorporated and key methodological factors that alleviate systematic biases were addressed, patellogastropods were classified as the earliest diverging gastropods, recovering the Orthogastropoda hypothesis (Uribe *et al.*, 2022). This review adopts the backbone for gastropods generated by Uribe *et al.* (2022) as previously accepted (Vortsepneva, Mikhlin & Kantor, 2023) (Fig. 3).

#### (b) *Patellogastropoda*

A monophyletic Patellogastropoda (true limpets) has been well supported by several autapomorphies (e.g. foliated shell structure and pairs of buccal cartilages) (Ponder & Lindberg, 1997) and molecular studies (Cunha & Giribet, 2019; Xu *et al.*, 2022; Qi *et al.*, 2023). However, relationships among limpets remain contradictory between traditional morphological studies and Sanger-derived sequences (e.g. Sasaki, 1998; Nakano & Ozawa, 2007). For instance, a clade uniting Nacellidae and Patellidae was supported by shared features such as pallial sensory organs, radular musculature, and odontophoral cartilages (Sasaki, 1998). However, this contradicts findings based on a multigene data set that Nacellidae is the sister taxon of Pectinodontidae + Lepetidae (Nakano & Ozawa, 2007), and even suggests non-monophyly of Nacellidae (Harasewych & McArthur, 2000; Nakano & Ozawa, 2007).

Although the majority of phylogenetic trees based on mitochondrial genes contain monophyletic families, topologies among families have remained incongruent (Xu *et al.*, 2022; Qi *et al.*, 2023) probably due to the loss of phylogenetic signal or differences in mutation rates across lineages (Uribe *et al.*, 2019). In these studies, Patellidae has consistently failed to form a sister-group relationship with Nacellidae. However, with a more expanded sampling of deep-sea taxa, the topologies among families remain consistent across all analyses, and the internal relationships of Patellogastropoda are nearly resolved and are in agreement with morphological assignments at family level (Ponder *et al.*, 2019; Qi *et al.*, 2023). To establish reliable relationships within Patellogastropoda, it will be not only essential to include sampling of rare taxa (e.g. Rhodopetalidae and other deep-sea families; Chen *et al.*, 2022), but also to increase the number of genes used in analyses. In addition, the higher classification of Patellogastropoda warrants further consideration.

#### (c) *Vetigastropoda s.s. + Neomphaliones*

Vetigastropoda is a connecting link between Patellogastropoda and the remaining gastropods, and is key to determining whether Orthogastropoda is monophyletic. It is unique in the presence of structures such as the widely distributed true sensory papillae (Ponder & Lindberg, 1997; Haszprunar *et al.*, 2017). This exclusively marine clade inhabits all latitudes and bathymetries (including hydrothermal vent habitats) and consists of a wide variety of forms among which are the abalone, keyhole limpets, top shells, turban shells, slit

shell snails, and numerous taxa with tiny shells (Salvini-Plawen, 1980; Geiger, Nützel & Sasaki, 2008; Ponder *et al.*, 2019). For a long time, the systematics of Vetigastropoda remained biased by homoplastic characters that were used to circumscribe lineages, e.g. pallial cavity symmetry (Uribe *et al.*, 2016), features of the shell and radula (Hickman & McLean, 1990), or physiological and morphological adaptations to hydrothermal habitats (Haszprunar, 1988a). Several molecular analyses have improved the systematics of Vetigastropoda more recently (Williams & Ozawa, 2006; Kano, 2008; Williams, Karube & Ozawa, 2008; Aktipis & Giribet, 2012; Donald *et al.*, 2012; Williams, 2012; Donald & Spencer, 2016; Uribe *et al.*, 2017b). However, multigene analyses and mitogenomics have proved insufficient for resolving the ancient divergences of Vetigastropoda due to limited evolutionary information (Kano, 2008; Aktipis & Giribet, 2012; Uribe *et al.*, 2016, 2017a; Guo *et al.*, 2021) and the widespread issue of long branch attraction (LBA) (Aktipis & Giribet, 2012; Guo *et al.*, 2021).

Cocculinids and neomphalids were initially excluded from Vetigastropoda due to the absence of epipodial sense organs (ESOs) (Salvini-Plawen & Haszprunar, 1987). However, later studies revealed that adult Cocculinidae do have simple ESOs, reflecting either paedomorphosis or a plesiomorphic condition (Haszprunar *et al.*, 2017). Alternatively, Cocculinida and Neomphalida were grouped together as the only two orders within the subclass Neomphaliones (Bouchet *et al.*, 2017). However, the latest classification includes them as suborders within Neomphalida (Ponder *et al.*, 2019), and this position was supported in a recent phylogenomic study (Uribe *et al.*, 2022). Consequently, Pleurotomariida + Neomphaliones (comprising Cocculinida and Neomphalida) was recovered as a sister clade to the remaining vetigastropods (Haliotida, Seguenziida, Lepetellida, Fissurellida, Trochida) (Uribe *et al.*, 2022) (Fig. 3). This refined understanding of Vetigastropoda supports the assumption that ESOs might be present in early juveniles of neomphalids and pleurotomariids (Haszprunar *et al.*, 2017). However, this taxonomy is not compatible with distribution of true epipodial tentacles. Branching second, the order Seguenziida was well resolved as a sister clade to the remaining orders, however, it is important to note that Seguenziida is still underrepresented with just one of its 10 families sampled. Together, Vetigastropoda *s.l.* is composed of Neomphaliones and Vetigastropoda *s.s.*, the latter includes Pleurotomariida, Haliotida, Seguenziida, Lepetellida, Fissurellida, Trochida.

Within Vetigastropoda *s.s.*, conflicts remain regarding the monophyly of Fissurellida, as well as the relationships among Lepetellida *s.s.*, Haliotida, and Trochida (Cunha *et al.*, 2022; Fig. 3). A better taxon sampling of under-represented lineages (mainly from deep-sea and small-sized snails) will be necessary to provide a reliable classification of the diversity of these lineages.

#### (d) *Adenogonogastropoda (=Angiogastropoda)*

Adenogonogastropoda is a clade containing Neritimorpha + Apogastropoda (Heterobranchia + Caenogastropoda), and

comprises the greatest diversity of the gastropods with nearly 90,000 living species (Bánki *et al.*, 2023). The association of these subclasses was first proposed by Simone (2011) based on developmental and anatomical synapomorphies (e.g. larva condition, vision using eyes, and a pallial reproductive duct) and then by Cunha & Giribet (2019) based on the enclosed nature of the embryo after internal fertilization (as Angiogastropoda). The number of radular teeth in each transverse row can generally be used to distinguish between Neritimorpha (rhipidoglossan radula with numerous teeth) and Apogastropoda (mostly taenioglossan radula with seven teeth) (Vortsepneva *et al.*, 2023). In addition to morphological studies, mitogenomic (Uribe *et al.*, 2019) and phylogenomic analyses (Kocot *et al.*, 2011; Zapata *et al.*, 2014; Cunha & Giribet, 2019; Uribe *et al.*, 2022) also support Adenogonogastropoda.

Despite having a relatively low number of species, Neritimorpha (with a single order; Fig. 3) contains great morphological variety (from typical snail shapes, to limpet-like, and even shell-less taxa) adapted to numerous habitats (from the deep sea, including hydrothermal vents, to shallow rocky shores, freshwater, land, and even taxa inhabiting trees) (Lindberg, 2008). This subclass has consistently been recovered as monophyletic (Cunha & Giribet, 2019; Uribe *et al.*, 2022). The first phylogenetic tree of Neritimorpha potentially included stochastic error or LBA, because it used only a partial gene (28S rRNA) and Hydrocenoidea displayed a long branch (Kano, Chiba & Kase, 2002). Later studies using mitogenomic and nuclear data sets recovered (Neritopsoida (Helicinoidea (Neritoidea + Hydrocenoidea))) (Uribe *et al.*, 2016). This same structure was recovered by phylogenomic analyses (Cunha & Giribet, 2019; Uribe *et al.*, 2022), although Hydrocenoidea was not included in these analyses.

#### (e) *Apogastropoda*

Apogastropoda contains Heterobranchia and Caenogastropoda, the two most diversified subclasses. This clade was initially proposed by Salvini-Plawen & Haszprunar (1987) and was finally defined as monophyletic by Ponder & Lindberg (1997) based mostly on sperm synapomorphies. In addition, Apogastropoda has been recovered in a broad range of morphological (Simone, 2011) and molecular phylogenies using multigene (Colgan *et al.*, 2003), mitogenomic (Castro & Colgan, 2010; Uribe *et al.*, 2019), and genome-scale data (Kocot *et al.*, 2011; Zapata *et al.*, 2014; Cunha & Giribet, 2019; Uribe *et al.*, 2022). It is possible that, as our understanding of the fossil record improves, the presumed sister relationship between caenogastropods and heterobranchs could potentially collapse. Other extinct non-heterobranch taxa, such as the extinct neritimorphs, may then be identified as the sister group of caenogastropods. This is possible considering that Late Paleozoic neritimorphs possess homeostrophic larval shells similar to those of caenogastropods (Nützel *et al.*, 2007), and both groups are inferred

to have arisen during the Ordovician period (Ponder *et al.*, 2019).

#### (f) *Heterobranchia*

Heterobranchia is a highly diversified lineage encompassing a wide variety of snails and slugs adapted to marine, freshwater, and terrestrial environments (Moles & Giribet, 2021). Historically, it comprised Euthyneura (Opisthobranchia and Pulmonata) and the 'lower Heterobranchia' (=Allogastropoda) (Haszprunar, 1985; Ponder & Lindberg, 1997). Although Heterobranchia are characterised by several synapomorphies, such as a hypertrophic protoconch, spiral sperm, and egg-strings, these features are less useful for defining its subgroups (Haszprunar, 1985). Most relationships among major heterobranch groups remain controversial across molecular investigations to date. Among these groups, 'lower heterobranchs' are critical to understanding heterobranch evolution, many of which contain minute or microscopic groups (e.g. Valvatoidea, Orbitestelloidea, Omalogyroidea, and Rissoelloidea) (Dinapoli & Klussmann-Kolb, 2010; Ponder *et al.*, 2019; Varney *et al.*, 2021).

Mid-Paleozoic 'lower heterobranchs' pre-date Jurassic Euthyneura in the fossil record (Bandel, 1996; Frýda, Nützel & Wagner, 2008; Gründel & Nützel, 2012). The earliest branching of 'lower heterobranchs' is also supported by most phylogenetic analyses, but the details of their relationships remain inconsistent (Mordan & Wade, 2008; Jörger *et al.*, 2010; Wägele *et al.*, 2014; Kano *et al.*, 2016; Ponder *et al.*, 2019; Varney *et al.*, 2021). For example, Valvatoidea and Orbitestelloidea are sometimes recovered as a polytomy, a finding that contradicts the distinct progression of Valvatoidea towards other heterobranchs, as inferred from some molecular data and morphological features like the typical taenioglossan radula and specialised genital system (Haszprunar, 1988b; Varney *et al.*, 2021). Meanwhile, Pyramidellidae and Rissoellidae previously considered to be members of the 'lower heterobranchs' are now included in Euthyneura (Dinapoli & Klussmann-Kolb, 2010; Kano *et al.*, 2016; Varney *et al.*, 2021), a change supported by similarities in sperm morphology (Healy, 1988). Thus, whether 'Lower Heterobranchia' constitutes a clade sister to all remaining heterobranchs or the first branching paraphyletic grade remains in need of additional research.

Within Euthyneura, there has been broad sampling to test the monophyly of Opisthobranchia, Panpulmonata, and Pulmonata (Dinapoli & Klussmann-Kolb, 2010; Varney *et al.*, 2021), and efforts in phylogenomics (Kocot *et al.*, 2013). Opisthobranchia is generally considered to be paraphyletic, and the monophyly of Pulmonata has not been consistently supported (Kocot *et al.*, 2013; Varney *et al.*, 2021). Although monophyly of Panpulmonata was supported by partial genetic markers (Dinapoli & Klussmann-Kolb, 2010), mitogenome-based Bayesian inference (Varney *et al.*, 2021), and phylogenomics (Kocot *et al.*, 2013), the positions of Siphonarioidea and *Sacoglossa* remain unstable (Kocot *et al.*, 2013; Varney *et al.*, 2021). Such incongruences hinder

our understanding of gastropod radiation and the transition from a plicate gill to pulmonate lung (Fig. 3). Further expansion of genomic and taxon sampling is still needed, along with a comprehensive revision of heterobranch systematics (Kocot *et al.*, 2013; Ponder *et al.*, 2019).

#### (g) *Caenogastropoda*

As the most diverse gastropod lineage, Caenogastropoda comprises over 150 living families (Bouchet *et al.*, 2017). This successful taxon has been confirmed as monophyletic based on both morphological (e.g. osphradial cells and sperm ultra-structure) and molecular phylogenetics (Haszprunar, 1985, 1988b; Ponder & Lindberg, 1997; Simone, 2011; Zou, Li & Kong, 2011; Cunha & Giribet, 2019; Li *et al.*, 2024). Architaenioglossa, Sorbeoconcha (Caenogastropoda except architaenioglossans), and Hypsogastropoda (which have an exophalic penis) are major caenogastropod groups (Ponder & Lindberg, 1997; Colgan, Ponder & Eggler, 2000; Ponder *et al.*, 2019). Ponder *et al.* (2019) revised the classification of Architaenioglossa, incorporating Campanilida into the group. However, the monophyly of Architaenioglossa remains unstable across morphological and molecular phylogenies (Simone, 2011; Osca, Templado & Zardoya, 2015; Li *et al.*, 2024). Although the monophyly of both Sorbeoconcha and Hypsogastropoda is always supported by morphological and molecular data (Simone, 2011; Osca *et al.*, 2015; Irwin *et al.*, 2021; Li *et al.*, 2024), the relationships within Hypsogastropoda remain obscure to date (Fig. 3). This uncertainty may be attributed to a rapid radiation following the Permian–Triassic extinction or to heterochrony (Ponder & Lindberg, 1997; Colgan *et al.*, 2007), which also hinders further exploration of the predicted whole-genome duplication (WGD) event within this group (Hallinan & Lindberg, 2011). For example, the circumscription of Neogastropoda remains unclear due to the unstable position of Tonnoidea and Cancellarioidea with mitogenomic data (Osca *et al.*, 2015; Irwin *et al.*, 2021; Lemarcis *et al.*, 2022; Li *et al.*, 2024; Wang *et al.*, 2024a) and phylogenomics (Fedosov *et al.*, 2024). The term ‘Littorinimorpha’ is used informally to compile a group of non-neogastropod hypsogastropods (Ponder *et al.*, 2019). Unfortunately, the detailed relationships within this group remain largely in flux, including: (i) paraphyletic Littorinimorpha (Cunha & Giribet, 2019); (ii) the contested monophyly of Tonnoidea (Colgan *et al.*, 2007; Zou *et al.*, 2011); and (iii) relationships among Vanikoroidea, Truncatelloidea and Rissooidea (Criscione & Ponder, 2013; Takano & Kano, 2014; Criscione *et al.*, 2017). The predicted WGD event within Hypsogastropoda, with a very high posterior probability, is on a branch containing the Neogastropoda and some littorinimorphan families (Hallinan & Lindberg, 2011).

To our knowledge, only a few phylogenetic studies have been dedicated to elucidating the phylogeny of Caenogastropoda as a whole (Colgan *et al.*, 2007; Simone, 2011; Zou *et al.*, 2011; Osca *et al.*, 2015; Irwin *et al.*, 2021; Li *et al.*, 2024). Despite relatively extensive taxon sampling, the mitogenomic gene loci lack

a sufficient phylogenetic signal. It is likely that a more reliable molecular systematics of caenogastropods will be achieved following genome-scale analyses. In this context Cunha & Giribet (2019) made some progress, but with limited caenogastropod sampling. Thus, conducting robust phylogenomic studies within this group, combined with more high-quality genomes and increased taxon sampling, will enable the detection of WGD events within Hypsogastropoda and shed light on the mechanisms driving species diversification in this highly diverse gastropod group (Pardos-Blas *et al.*, 2021; Herráez-Pérez *et al.*, 2023).

### (3) Other conchiferan lineages

#### (a) *Cephalopoda*

Cephalopoda represents the third largest molluscan class. They are diverse, but can be characterised by unique features such as a set of tentacles used for manipulation and sensory perception, horny beaks for feeding, and complex eyes providing advanced visual capabilities. It contains two extant subclasses: Nautiloidea and Coleoidea (Fig. 3). Of these, Nautilidae is the sole extant representative of the Nautiloidea, characterised by an external, chambered shell and pin-hole eye. The Coleoidea are characterised by having an internalised and reduced shell, or in some cases, a completely absent shell (Lindgren, Giribet & Nishiguchi, 2004). A sister relationship between the two major coleoid clades – the superorders Decapodiformes and Octopodiformes – is well supported not only by the number of tentacles, but also by mitogenomic (Strugnell *et al.*, 2017; Uribe & Zardoya, 2017; Tang *et al.*, 2019, 2021) and phylogenomic studies (Kocot *et al.*, 2011; Smith *et al.*, 2011; Tanner *et al.*, 2017; Anderson & Lindgren, 2021; Lindgren *et al.*, 2023).

Relationships within Coleoidea have long been a subject of debate (Allcock, Lindgren & Strugnell, 2015), particularly those within the Decapodiformes. For instance, a clade uniting Oegopsida, Spirulida, and Bathyteuthida has been supported based on mitogenomic studies (Strugnell *et al.*, 2017; Tang *et al.*, 2019, 2021) and the absence of the cornea and circularis muscle in their suckers (Young, Vecchione & Donovan, 1998; Vecchione, Young & Carlini 2000). However, the relationships among these three orders remain unresolved (Fig. 3). As another example, despite the fact that both extant Spirulida and Sepiidae (Sepiida) possess a similar phragmocone (a gladius-like chambered shell) (Strugnell *et al.*, 2017), a close relationship has not been confirmed by recent phylogenomic studies (Anderson & Lindgren, 2021; Lindgren *et al.*, 2023). Notably, early spirulid fossils exhibit oblique septa similar to those in Sepiida (Fuchs & Košťák, 2016). Additionally, the internal protoconch morphology of *Conoteuthis* (an extinct group of belemnites) rejects the presumed origin of early sepids from the Cretaceous *Conoteuthis* (Fuchs, Keupp & Wiese, 2012). Fossil studies of ancestral coleoids combined with phylogenomics could help to elucidate the evolutionary relationships between Spirulida and Sepiida.

(b) *Scaphopoda and Monoplacophora*

Despite the number of extant species in Scaphopoda (tusk shells) being approximately 15–20 times greater than that of Monoplacophora (Tryblidia), with around 500–600 and about 30 species respectively, both these classes are among the least diverse within the conchiferan molluscs. Within Scaphopoda, both morphological data (e.g. shape of the shell and the lateral tooth base; Steiner, 1998) and molecular analyses (Steiner & Dreyer, 2003; Steiner & Reynolds, 2003; Song *et al.*, 2023) have confirmed a sister relationship between the orders Dentaliida and Gadilida. The recent Monoplacophora consisting only of Neopilinida, exhibit rather little phenotypic variation, and have been regarded as ‘living-fossil’ group (Kano *et al.*, 2012). All three orders have strong support for their respective monophyly, but resolution below that level varies significantly.

Notably, several challenges exist. The classification of the earliest scaphopod fossils is complicated due to misleading features (Reynolds & Steiner, 2008). Additionally, the fossil record of extinct monoplacophorans has a gap from the Devonian to the boundary of the Neogene and Quaternary epochs (Taviani, Sabelli & Candini, 1990; Warén & Gofas, 1996). It is possible that deep-sea deposits could bridge this gap in the fossil record (Kano *et al.*, 2012). These shortcomings hamper the interpretation of their phylogeny, especially in the absence of molecular data (Kano *et al.*, 2012). Integrating the molecular clock not only will allow us to assess their divergence more accurately but also assist in filling in gaps in the fossil record.

Until recently, there were few molecular studies reconstructing the phylogeny of Scaphopoda (Haszprunar, 2008) and Monoplacophora (Reynolds & Steiner, 2008; Kano *et al.*, 2012). Only a few representatives of these two lineages have undergone transcriptomic or genome-scale sequencing (Kocot *et al.*, 2011, 2020; Smith *et al.*, 2011; Song *et al.*, 2023). The available molecular data are still largely limited to a few nuclear and mitochondrial genes and a small number of living representatives (Steiner & Dreyer, 2003; Steiner & Reynolds, 2003; Giribet *et al.*, 2006; Kano *et al.*, 2012).

## (4) Relationships within aculiferan lineages

(a) *Polyplacophora*

Polyplacophora, also known as chitons, are generally hypothesised to have a basal position in Mollusca. Therefore, their configuration is assumed to be plesiomorphic, providing a better interpretation for the secondary reduction of the dorsal shell plates in aplacophorans and the derived conchiferans (Vinther *et al.*, 2012; Vinther, 2015). Valve characters, egg hull type, sperm ultrastructure, and gill placement all feature in chiton classification (Sirenko, 1993, 1997; Buckland-Nicks, 1995; Kaas, Jones & Gowlett-Holmes, 1998). *Callochiton* (Fig. 1C) was previously classified under Chitonida due to its adanal type of gills (Sirenko, 2006), but may instead represent the sister group to Chitonida,

supported by both sperm and egg characters and the fertilization process (Buckland-Nicks & Hodgson, 2000; Buckland-Nicks, 2008) as well as molecular evidence (Sigwart *et al.*, 2013; Irisarri *et al.*, 2020; Liu, Sigwart & Sun, 2023). Its systematic position remains equivocal (Buckland-Nicks, 1995; Okusu *et al.*, 2003). Recently, the order Callochitonida, containing Callochitonidae only, was erected by Giribet & Edgecombe (2020). Therefore, extant chitons currently are divided into three well-resolved orders: Lepidopleurida, Callochitonida, and Chitonida (Irisarri *et al.*, 2020; Moles *et al.*, 2021; Liu *et al.*, 2023).

(b) *Aplacophora (Solenogastres + Caudofoveata)*

Aplacophora (Solenogastres + Caudofoveata) possess vermiform bodies and a shiny appearance, due to the possession of a cuticle and aragonitic sclerites. Solenogastres are generally characterised by a ciliated ventral groove, a unique sensory region (atrial sense organ), a muscular foregut bearing a radula, a midgut without compartments, and a posterior palliai cavity that lacks ctenidia. Caudofoveata do not have a ventral groove, and the area anterior to the mouth is surrounded by an oral shield. Their foregut is similar to that of Solenogastres, but the midgut is divided into a duct and sac. In contrast to Solenogastres, Caudofoveata possess a pair of true ctenidia in the posterior palliai cavity. Molecular data recovered the reciprocal monophyly of Solenogastres and Caudofoveata with strong support (Kocot *et al.*, 2019).

In early studies, hard parts, such as radulae and spicule characters, were considered effective for constructing the phylogeny of aplacophorans (Scheltema & Schander, 2000). Within Solenogastres, four orders were initially recognised based on morphology of the cuticle, sclerites, and lateroventral foregut glands: Pholidoskepia, Neomeniamorpha, Sterrofulstia, and Cavibelonia (Salvini-Plawen, 1978). Of these, the monophyly of Cavibelonia and Pholidoskepia was supported by morphological characters (Salvini-Plawen, 2003, 2004), but not by phylogenomics. In phylogenomics studies, Cavibelonia appeared as polyphyletic, indicating parallel evolution of hollow sclerites in multiple lineages (Kocot *et al.*, 2019). Despite strong support for the monophyly of Pholidoskepia, placement of Macellomeniidae and Meiomeniidae sometimes showed weak support (Kocot *et al.*, 2019). In addition, the status of Sterrofulstia has remained unclear in phylogenomics studies and due to extensive morphological variation in Cavibelonia (Kocot *et al.*, 2019). Thus, the traditionally recognised higher-level taxon of Solenogastres is not consistent with current molecular evidence.

Within Caudofoveata, three families are recognised: Prochaetodermatidae (Salvini-Plawen, 1975), Chaetodermatidae (Ihering, 1876), and Limifossoridae (Salvini-Plawen, 1970) based on morphology of the radula and the shape of the body and oral shield. Their phylogenetic relationships have been investigated using molecular data in the past 5 years (Mikkelsen, Kocot & Halanych, 2018; Kocot *et al.*, 2019; Mikkelsen *et al.*, 2019). Traditionally, Limifossoridae was viewed

as the first caudofoveate offshoot based on a putatively plesiomorphic distichous radula and simple body pattern (Salvini-Plawen, 1977). However, recent analyses, based on a few genes or on phylogenomics, placed Prochaetodermatidae consistently as sister to all other Caudofoveata (Mikkelsen *et al.*, 2018, 2019; Kocot *et al.*, 2019). This placement is supported by the unique prochaetodermatid radula, a pair of cuticular jaws, and some features of the digestive system (Mikkelsen *et al.*, 2018; Kocot *et al.*, 2019). Notably, the central chitinous plate of the prochaetodermatid radula is morphologically similar to the ancestral molluscan or aplacophoran radula (a rhachidian tooth), as inferred from fossils (Smith, 2012; Scheltema, 2014). This may indicate homology, and provide further evidence, in line with molecular data, for an early branching of Prochaetodermatidae within Caudofoveata (Mikkelsen *et al.*, 2018). Chaetodermatidae is either nested within Limifossoridae (Mikkelsen *et al.*, 2018, 2019), or is a sister taxon to Limifossoridae (Kocot *et al.*, 2019), with their true relationships remaining uncertain.

#### IV. MICROMOLLUSCS: ENIGMATIC AND OVERLOOKED BIOTAS

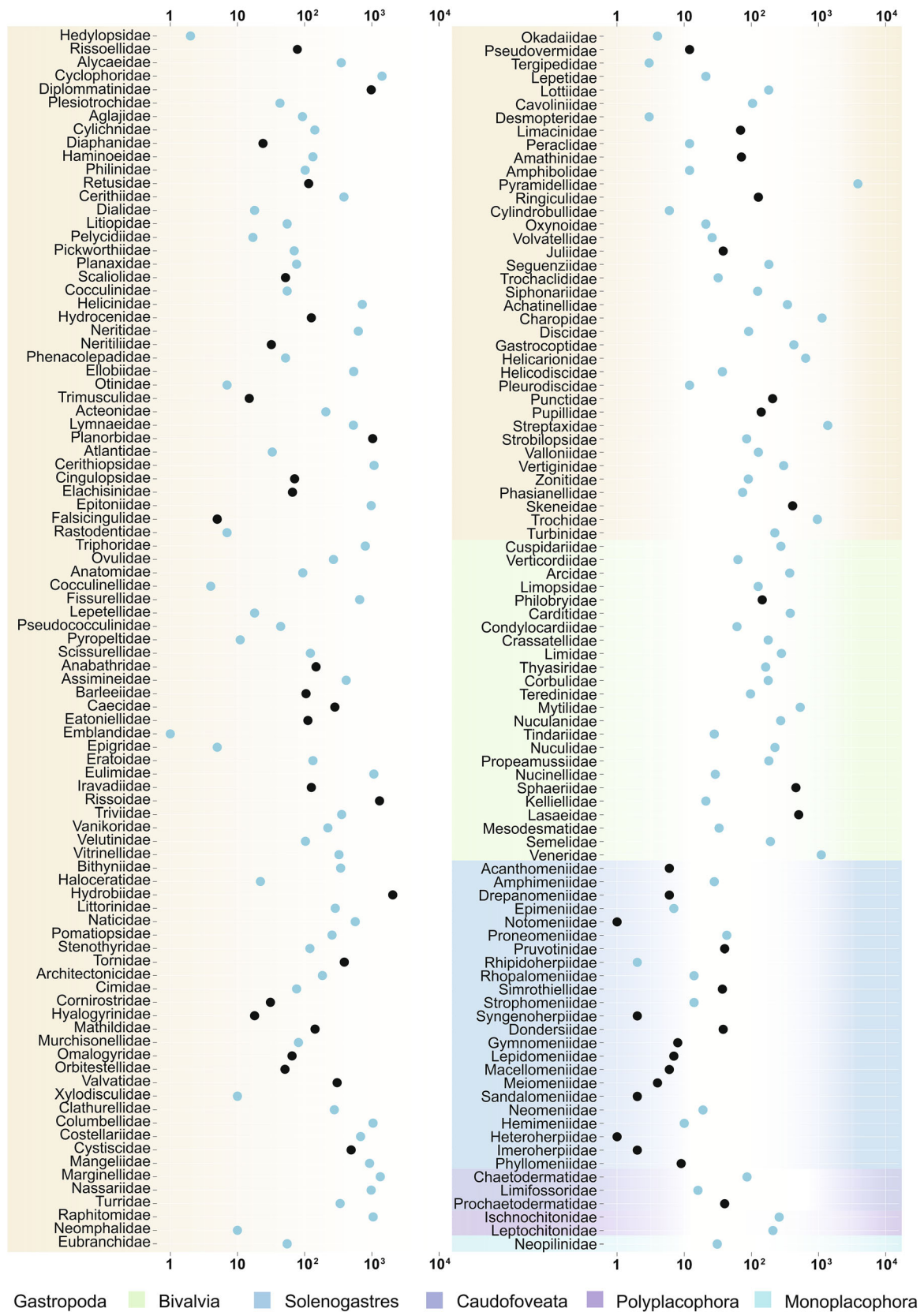
Micromolluscs are molluscs that require special magnification devices, such as microscopes or magnifiers, for observation (Middelfart *et al.*, 2016). There is some uncertainty in the definition of the maximum size of micromolluscs, ranging from 2 to 10 mm among studies (Kay & Kawamoto, 1983; Hasegawa, 1995; Hicks, Barrera & Tunnell, 2001; Bouchet *et al.*, 2002; Narciso, Prieto-Arcas & Acosta-Balbas, 2005; Sasaki, 2008). Herein, we use the widely accepted definition of 5 mm (Geiger *et al.*, 2007; Aravind, Patil & Madhyastha, 2008; Hickman, 2008; Durkan *et al.*, 2013; Queiroz & Dias, 2014; Nekhaev & Krol, 2020).

More than 180 family-level taxonomic units related to micromolluscs have been identified to date (Sasaki, 2008; Fig. 5). The majority of micromolluscs are gastropods or bivalves, with smaller representations in Aplacophora (AplacBase, <https://aplacbase.weebly.com/>), Polyplacophora (Saito, 1997), and Monoplacophora (Sasaki, 2008) (Fig. 5). These organisms hold great potential for assessing marine biodiversity (Olabarria & Chapman, 2001) and play vital roles in species conservation and ecological monitoring (Nekhaev & Krol, 2020; Zenetos *et al.*, 2020; Albano *et al.*, 2021). Micromolluscs are widely distributed, and occupy various ecological niches similar to macroscopic molluscs. They are found globally, in terrestrial and freshwater environments as well as marine habitats ranging from intertidal zones to shallow seas. Since the 1990s, our understanding of their habitats has expanded. They have also been discovered in unique habitats such as the mesopelagic to abyssal zones (Bouchet & Warén, 1993), sea-floor caves (Hayami & Kase, 1993), sunken wood (Hasegawa, 1997), and hydrothermal vents (Okutani, Fujikura & Sasaki, 1993; Warén & Bouchet, 1993).

Incorporating micromolluscs into molecular systematics could potentially yield new insights and refine our understanding of the evolution of major molluscan lineages, despite the challenges that it presents. For instance, small shelly fossils (SSFs) from the Late Precambrian and Cambrian periods are characterised by a rich diversity of diminutive (0.5–5 mm) organisms (Parkhaev, 2008). Monoplacophora, many of which are smaller than 3 mm (Sasaki, 2008), occupies a key node in the diversification of Conchifera. By integrating information on extant monoplacophorans, such as *Laevipilina antarctica* (3.2 mm body size) and *Laevipilina hyaline* (2 mm body size), future research will not only be able to classify molluscan SSFs better, but also to advance our understanding of the early branching patterns of conchiferans (Smith *et al.*, 2011; Kocot *et al.*, 2020). In addition, many miniaturized molluscan taxa exhibit faster rates of evolution, as has been observed in some rapidly evolving microgastropods today, including *Omalogyra atomus* (Omalogyroidea) and *Rissoella morrocanensis* (Rissoelloidea) (Varney *et al.*, 2021). Both Omalogyroidea and Orbitestelloidea consist almost entirely of microscopic species and have been regarded as important taxa for elucidating the early evolution of Heterobranchia (Varney *et al.*, 2021).

In some phylogenomic studies, micro-sized species are yet to be fully sampled. Examples include the unsampled Acanthomeniidae in Aplacophora (Kocot *et al.*, 2019) and the incompletely represented Lepetellida in Vetigastropoda (Cunha *et al.*, 2022). Moreover, within Littorinoidea, a recent study suggested that increased sampling of microgastropods can challenge previous hypotheses based on morphological and anatomical evidence (Qi *et al.*, 2024), in this case with the proposal that *Peasiella* and *Lacuna* may not belong to the Littorinidae, and a potential ancestral origin for *Pomatias* outside of the Littorinoidea. Thus, incorporating micromolluscs into molluscan systematics not only fills sampling gaps and expands the taxonomic range covered, but can also target key evolutionary nodes. This strategy may be particularly effective for groups composed primarily of micro-sized species. Several micromollusc groups may be key to understanding evolution within major groups or essential for constructing a robust phylogeny, including: (i) Acanthomeniidae (Solenogastres); (ii) Prochaetodermatidae (Caudofoveata); (iii) Neolepetopsidae (Patellogastropoda); (iv) Lepetellida (Vetigastropoda); and (v) Valvatoidea and Orbitestelloidea (Heterobranchia).

However, it is important to note that only limited molecular data are available for micromolluscs and they exhibit variable quality (Middelfart *et al.*, 2016). Many micromollusc species have not undergone DNA sequencing, and previous attempts to resolve the molecular phylogeny of micromolluscs have relied on small subsets of genes from a few taxa, such as Cingulopsioidea and Rissooidea (Criscione & Ponder, 2013; Criscione *et al.*, 2017), Rissoelloidea (Siadén *et al.*, 2019), and Caudofoveata (Mikkelsen *et al.*, 2018, 2019). Additionally, while transcriptome and genomic sequencing data are available for some aplacophorans (Kocot *et al.*, 2019) and monoplacophorans (Smith *et al.*, 2011; Kocot *et al.*, 2020), whole-genome sequencing



**Fig. 5.** 182 families containing micromolluscs recorded to date. Dots represent the number of species recorded in corresponding families, with families composed entirely of micromolluscs in black.

combined with third-generation sequencing technologies represents a promising avenue to enhance our understanding of micromolluscs.

## V. FACTORS DRIVING INCONGRUENCES

Phylogenomics has significantly advanced our ability to elucidate the evolution of life at all levels (Kapli *et al.*, 2020; Steenwyk *et al.*, 2023). However, incongruences remain pervasive in resolving the tree of molluscs at various levels, including conflicting tree topologies within or across class levels. These are constrained by a variety of factors, including biological factors and analytical errors (Steenwyk *et al.*, 2023). Here we summarise the major factors leading to incongruences, in the hope of deepening our understanding of molluscan evolution.

### (1) Biological factors

#### (a) Incomplete lineage sorting

Incomplete lineage sorting (ILS) (the mismatch between gene trees and species trees due to random genetic inheritance) is a primary contributor to current incongruences in deep molluscan relationships. Irrespective of computational costs and error rates, both fully and summary coalescent-based methods can be implemented to estimate ILS-caused incongruence. The latter was applied to detect incongruence among major molluscan clades within Conchifera (Cephalopoda, Gastropoda, Scaphopoda, and Bivalvia) and showed that ILS may be responsible for the uncertainty of scaphopod placement (Song *et al.*, 2023). ILS is more likely to be detectable in cases of recent divergence. However, molluscan genomic studies indicate the occurrence of ILS dating back approximately 500 million years, suggesting that molluscs may serve as a useful model for exploring ILS in ancient radiation episodes, especially the Cambrian Explosion (Song *et al.*, 2023).

#### (b) Whole-genome duplication

Whole-genome duplications (WGDs) have been documented in *Achatina immaculata* and *A. fulica* (two giant African snails; Liu *et al.*, 2021a), Mytilidae (Corrochano-Fraile *et al.*, 2022), and Conidae (Pardos-Blas *et al.*, 2021). Detection of WGDs can provide deeper insights into the genetic mechanisms underpinning the diversity of molluscs and their different body plans. However, incomplete genome data and methodological challenges hinder our ability to detect WGDs and duplicated genes in Mollusca (Liu *et al.*, 2021a; Corrochano-Fraile *et al.*, 2022).

#### (c) HGT, hybridization, and natural selection

Aside from ILS and WGD, other biological factors such as horizontal gene transfer (HGT), hybridization, and natural selection could account for incongruences in molluscan

phylogenies at all levels, but are less studied. The establishment of intimate host–microbe relationships can dramatically increase the chances of HGT (Husnik & McCutcheon, 2018). However, HGT from bacteria to molluscs has been detected only in some species of cephalopods (Liu *et al.*, 2018), gastropods (Rumpho *et al.*, 2008), and bivalves (Ip *et al.*, 2021). It is still unclear whether HGT is pervasive in molluscs. Hybridization and natural selection also can obfuscate the phylogenetic placement of taxa. For example, a genome-wide study suggested two independent origins of live-bearing in the genus *Littorina* (Stankowski *et al.*, 2024). This discordance between evolutionary relationships and reproductive modes could result from an interplay between hybridization and selection on specific sets of alleles (Stankowski *et al.*, 2024).

### (2) Analytical factors

#### (a) Stochastic errors

Taxon sampling and locus sampling are two sources of stochastic errors of incongruence when inferring a molluscan tree. Numerous efforts have been made regarding these two aspects in an attempt to reconstruct relationships among or within molluscan clades (Zapata *et al.*, 2014; González *et al.*, 2015; Kocot *et al.*, 2019, 2020; Lemer *et al.*, 2019; Uribe *et al.*, 2022; Song *et al.*, 2023). For example, in studies attempting to elucidate the evolutionary relationships among the major molluscan classes, different species-sampling strategies caused incongruence in the phylogenetic placement of Gastropoda, Scaphopoda, and Bivalvia (Kocot *et al.*, 2011, 2020; Song *et al.*, 2023).

Within classes, use of mitogenomics and phylotranscriptomics has supported a consistent sister relationship between Apogastropoda (Caenogastropoda + Heterobranchia) and Neritimorpha, but conclusions regarding the earliest divergences within gastropods remain in conflict (Kocot *et al.*, 2011; Zapata *et al.*, 2014; Cunha & Giribet, 2019; Uribe *et al.*, 2022). Incorporating under-sampled deep-sea taxa could significantly advance our understanding of the backbone of the major molluscan classes (e.g. Uribe *et al.*, 2022; González-Delgado *et al.*, 2024). Taxon occupancy thresholds per locus of around 50% are commonly applied in phylogenomic research (Shen *et al.*, 2018; Steenwyk *et al.*, 2019) despite it being known that these data sets are more susceptible to systematic errors, even resulting in highly supported erroneous topologies within specific molluscan clades (Song *et al.*, 2023). A low threshold could negatively impact tree topology due to missing data or non-random distributions of missing data (Simmons, 2012; Uribe *et al.*, 2022; Steenwyk *et al.*, 2023).

#### (b) Systematic errors

Systematic errors and treatment errors also can lead to incongruences (Kapli *et al.*, 2020; Steenwyk *et al.*, 2023). Among possible systematic errors, branch-length heterogeneity and compositional bias are the two issues of primary concern in molluscan phylogenomics (Kocot *et al.*, 2017).

The concatenation-based method is commonly used to infer a single molluscan phylogenomic tree. However, concatenation-based and coalescent-aware methods can yield conflicting results when constructing deep molluscan topologies (e.g. Uribe *et al.*, 2022; Song *et al.*, 2023). Deciding on the most suitable approach for curated molluscan data sets can be challenging.

## VI. WHAT TO DO NEXT?

### (1) Identifying key factors

While it is recognised that pinpointing sources of incongruence can be challenging, it is possible to make some judgments based on *a priori* knowledge (Kocot *et al.*, 2017; Steenwyk *et al.*, 2023). When attempting to elucidate deep divergences among molluscs at the class level, ILS can currently be considered a primary biological factor. Notably, in deep radiations, the detection of ILS can be difficult due to the feeble phylogenetic signal caused by substitutional saturation at the terminals (Kapli *et al.*, 2020). For the detection of systematic errors, five important factors – branch-length heterogeneity and saturation, amino acid compositional heterogeneity and overall evolutionary rate, and missing data – should be taken into consideration when constructing molluscan phylogenies (Kocot *et al.*, 2017). When conflicts in the placement of subclasses are observed, heterogeneity can serve as the main factor for subsequent analysis. For deeper nodes within a subclass, insufficient loci, sparse taxon sampling, and rapid divergence are likely the main factors contributing to incongruence (Cunha *et al.*, 2022).

### (2) Accommodating heterogeneity of rates

For taxa that have undergone multiple substitutions, high evolutionary rates are prone to causing LBA artefacts (Phillips, Delsuc & Penny, 2004). LBA is pervasive in some early-diverging molluscs, such as Patellogastropoda, Vetigastropoda, and ‘lower Heterobranchia’ (Varney *et al.*, 2021; Uribe *et al.*, 2022; Xu *et al.*, 2022). Strategies such as removing rapidly evolving sites/taxa (Cunha & Giribet, 2019; Varney *et al.*, 2021; Uribe *et al.*, 2022), use of mixture models together with amino acid recoding (Uribe *et al.*, 2019), and the addition of taxon/loci sampling (Varney *et al.*, 2021; Uribe *et al.*, 2022; Xu *et al.*, 2022) can enable the construction of more robust phylogenomic relationships. Additionally, reasonable out-group sampling can sometimes help mitigate LBA-related issues (Cunha & Giribet, 2019).

### (3) Handling compositional heterogeneity

To circumvent heterogeneous amino acid composition, matrix occupancy, discarding rapidly evolving sites, and amino acid recoding can be applied to infer ancient relationships within bivalve and gastropods (Lemer *et al.*, 2019; Uribe

*et al.*, 2022). Amino acid recoding can combat heterogeneity and improve the convergence of Bayesian Inference (BI) analysis to some extent (Cunha & Giribet, 2019). However, the commonly used six-state amino acid recoding can sometimes erode the phylogenetic signal (Lemer *et al.*, 2019; Uribe *et al.*, 2022). Use of a larger number of amino acid states (nine-state and 12-state) for comparison can enable this risk to be minimised (Hernandez & Ryan, 2021).

### (4) Data scale

In general, the accumulation of data (on taxa or genes) will facilitate a robust ToL. We suggest using high-quality whole-genome data in efforts to elucidate class-level relationships, while combining expanded taxon sampling and transcriptomics to address deep diversification within a class/subclass. Mitogenomes appear inadequate for determining a phylogeny at order level (e.g. in heterobranchs; Varney *et al.*, 2021), below the superfamily level [in chitons (Liu *et al.*, 2023) and neogastropods (Wang *et al.*, 2023)], and within rapidly evolving molluscan lineages. However, mitochondrial genomes or a number of Sanger loci have proved useful in molluscan phylogenetics within family and species levels, especially given the limited number of high-quality molluscan genomes available (Criscione *et al.*, 2017; Liu *et al.*, 2023; Qi *et al.*, 2023; Wang *et al.*, 2023). It is also worth considering increased out-group sampling when targeting the deep nodes of Mollusca (Cunha & Giribet, 2019). For researchers lacking the resources to sequence full genomes or transcriptomes, or those relying on use of older samples in laboratory and museum collections, ultraconserved elements (UCEs) offer a balance between dense sampling and increased loci. This method shows great promise for constructing relationships within a specific molluscan class at various taxonomic levels (González-Delgado *et al.*, 2024).

### (5) High-quality data set

High-quality, reliable data are essential for building a reliable molluscan ToL, especially in the era of phylogenomics. In order to ameliorate incongruences in deep-level relationships of molluscs caused by data quality, it is imperative to obtain a greater number of high-quality molluscan genomes that encompass the diversity of the group. Here, we propose that a ‘high-quality’ molluscan genome must meet the following criteria: (i) BUSCO (benchmarking universal single-copy orthologs) C (complete) score >90%, and D (duplicated) <5%; (ii) chromosome-scale assembly; and (iii) low assembly error, defined by Hi-C (high-throughput chromosome conformation capture) mis-orientation, low contamination, and a high mapping rate.

Notably, most available molluscan genomes are from taxa with unusual adaptations (Albertin *et al.*, 2015; Sun *et al.*, 2020; Ip *et al.*, 2021; Zhang *et al.*, 2021; Guo *et al.*, 2023) or economic importance (Zhang *et al.*, 2012; Wang *et al.*, 2017; Kenny *et al.*, 2020). Future phylogenomics analyses should prioritize taxon selection to address missing

genomic data (Ponder *et al.*, 2019; Sigwart *et al.*, 2021). Sigwart *et al.* (2021) identified several key taxa crucial to clarifying pan-molluscan phylogeny: (i) Nautilida; (ii) Lepidopleurida; (iii) Cycloneritida and Cocculinida; and (iv) Anomalodesmata and the protobranches. In addition to these groups, microscopic taxa nested within molluscs should also receive greater attention. Furthermore, when selecting a species as a representative example of a clade, genome size, GC content, and, if possible, heterozygosity are informative criteria (Sigwart *et al.*, 2021).

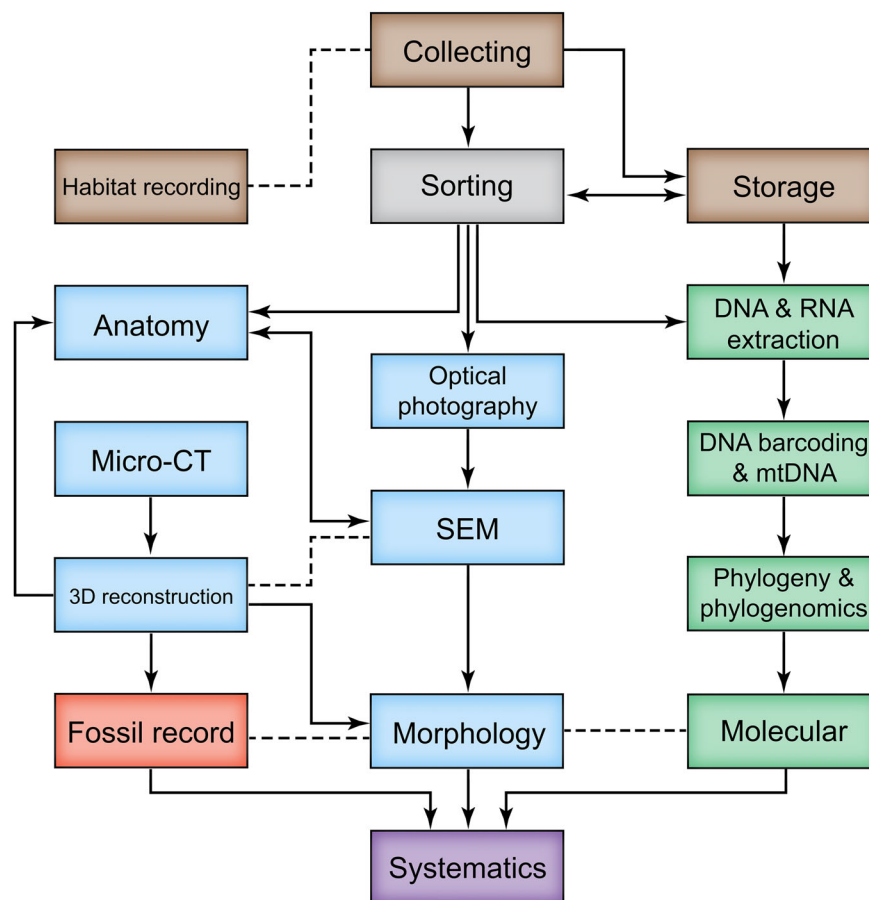
## (6) Data management

Considerable efforts have been made to curate databases such as MolluscDB (Caurcel *et al.*, 2021; Liu *et al.*, 2021b), providing a robust platform for the ongoing storage and management of mollusc genomic and transcriptomic data. This not only facilitates future research but also enables comprehensive analysis of existing data for molluscan systematics (Fig. 6). In addition to utilising phylogenomic approaches to

infer molluscan phylogeny, it will be crucial to establish mollusc sample repositories and distribution databases. This ensures a stable supply of samples, facilitating ecological adaptation studies. Simultaneously, there is a need to reassess the role of morphology. Utilising advanced techniques such as X-ray micro-computed tomography (micro-CT) scanning to examine the internal structures of extant taxa offers a non-destructive method to obtain and store high-resolution images (Ponder *et al.*, 2019) that can uncover detailed morphological features that are otherwise difficult to observe, enhancing our understanding of molluscan anatomy and evolution.

## (7) Evo-Devo clues

Future perspectives on molluscan phylogeny could benefit from incorporating evolutionary developmental biology (Evo-Devo) studies to provide deeper insights into the developmental processes and evolutionary mechanisms shaping



**Fig. 6.** Key steps and methods for molluscan systematics. In addition to expanded gene (green boxes) and taxon sampling, it is also useful to collect other types of information. Of these, recording the habitat and geographical locations of molluscs during fieldwork (brown boxes) is a crucial step in expanding the mollusc distribution database. For taxa that may be difficult to investigate using traditional anatomical methods, such as fossils (orange box) and micromolluscs, the application of morphological research techniques (blue boxes), including scanning electron microscopy (SEM) and micro-computed tomography (Micro-CT), enables the examination of both external and internal microscopic features. These features can further be integrated with phylogenomics to infer and interpret systematic relationships (purple box). mtDNA, mitochondrial DNA.

today's molluscan diversity. For example, it would be useful to investigate the roles of each type of chiton sclerite-secreting cell and the mechanisms modulating their specification and organization (Xia, Huan & Liu, 2023), as well as studying aplacophoran morphogenesis on both molecular and cellular levels (Kocot, 2013). Integration of Evo-Devo data can reveal how genetic and developmental pathways have evolved to contribute to the wide range of molluscan body plans.

## (8) Fossils

Besides data from living molluscs, incorporation of fossil taxa is equally important (Ponder *et al.*, 2019). Fossils play a crucial role in understanding molluscan phylogeny and evolution, providing a historical context that bridges the morphological gaps between extant and extinct forms (e.g. Vinther *et al.*, 2017; Zhang *et al.*, 2024). Fossils help in identifying ancestral traits and lineage-specific adaptations, shedding light on the evolutionary processes that have shaped the diverse forms of modern molluscs. The microstructure of fossils is taxonomically valuable and can provide clues for tracing the evolutionary history of characters (Sato *et al.*, 2020). By examining fossil records, we can infer the timing and sequence of key evolutionary innovations, such as the emergence of new body plans. Crucially, fossils contribute to constructing a nuanced timetree by providing critical calibration points for molecular clock analyses. A well-resolved phylogenomic framework for molluscan phylogeny will also help constrain high-level and deep relationships of recent and fossil taxa that are represented by only morphological characters, as often used in current palaeontological studies of early molluscan fossils (Zhang *et al.*, 2024). The hypothetical possibility of extracting DNA from million-year-old fossil molluscs is certainly exciting, but given that DNA is fragile and prone to contamination, it would undoubtedly be extremely challenging to achieve. Integrating morphological data from both extant and extinct taxa, Evo-Devo studies, and advanced phylogenomic approaches together will enable a more comprehensive and nuanced reconstruction of molluscan phylogeny, addressing current gaps and discrepancies in our knowledge (Fig. 6).

## (9) Chromosomal gene linkage

In addition to morphological characters and extensively sampled gene data, for the main classes of molluscs there are now chromosome-scale genome sequences available (see online Supporting Information, Table S1). This wealth of genome assemblies, together with the development of a suitable algorithm, has paved the way for synteny-based phylogenomics (Steenwyk & King, 2024). The use of chromosome-scale gene linkage as synteny-based characters, such as chromosomal translocations and fusion, could serve as a powerful tool for examining ancient syntenies and derived rearrangements across molluscs. This approach could significantly enhance our understanding of molluscan evolution (Song *et al.*, 2023; Sigwart

*et al.*, 2024) and has already been used effectively in a study of the earliest-diverging animals (Schultz *et al.*, 2023).

Taken together, building a robust molluscan phylogeny demands not only advanced bioinformatic methodologies and tools but also comprehensive sampling, high-quality genome-scale data, and the integration of multiple lines of evidence to capture the genetic and phenotypic diversity and evolution of molluscs. We argue here for a deep collaboration between molecular systematists and morphologists (Dayrat, 2005; Padial *et al.*, 2010; Sigwart *et al.*, 2021).

## VII. CONCLUSIONS

- (1) While notable progress has been made, our understanding of molluscan systematics remains incomplete. Future research must delve deeper into the phylogenetic relationships across and within major molluscan taxa, with a focus on expanding taxon sampling of lesser-known groups, often of small body sizes, and securing a greater amount of high-quality genomic data.
- (2) Micromolluscs and species from extreme environments could provide critical insights, potentially acting as key nodes for refining the molluscan Tree of Life.
- (3) In addition to molecular data, comprehensive data sets covering morphology, distribution and other ultrastructural information will undoubtedly continue to contribute to the ongoing improvement of our understanding of molluscan systematics. The integration of advanced technologies such as micro-computed tomography (micro-CT) and three-dimensional reconstruction, in conjunction with anatomical, developmental, and palaeontological studies, will be essential for the further advancement of this field. Efforts to secure stable funding, assemble skilled teams, and foster enhanced collaboration between molecular systematists and morphologists are crucial for propelling the field of molluscan systematics forward.

## VIII. ACKNOWLEDGEMENTS

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## X. SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.

**Table S1.** Summary data on chromosome-level genomes available across Mollusca.

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